

TRANSPORTATION TRENDS IN BARCELONA'S SUBURBAN AND PERI- URBAN ZONES



BEYOND BARCELONA'S HEART: ARE ITS METROPOLITAN CITIES SHIFTING FROM SUSTAINABLE TRAVEL TOWARDS CAR DEPENDENCY?

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INTRODUCTION

While urban studies emphasize city centers, this study examines whether sustainable travel habits extend to Barcelona's suburbs. Results offer insights for promoting sustainable transport across metropolitan regions.

OBJECTIVE

1. Analyze travel mode shifts from 2012 to 2018 in Barcelona's suburbs.
2. Assess if local accessibility supports active travel or if regional transit is needed to curb car dependency.

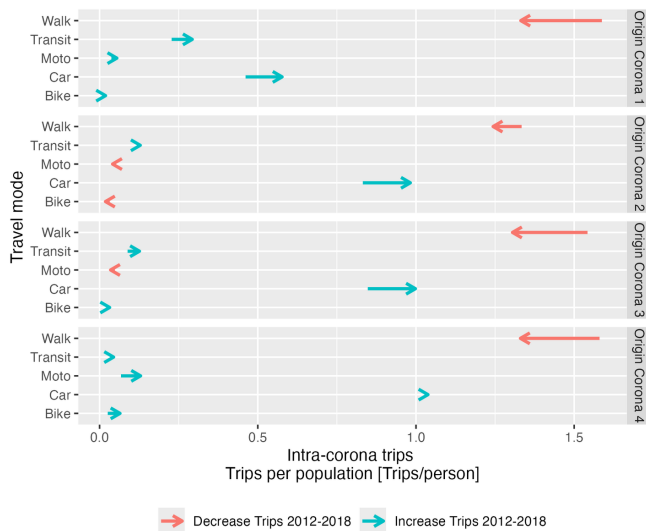
METHODOLOGY

We analyzed 2012-2018 travel trends across **four transit fare coronas** in Barcelona's metropolitan area using local travel survey, CORINE urbanization data, and transit supply metrics. This approach reveals shifts in travel modes, highlighting car dependency and accessibility patterns.

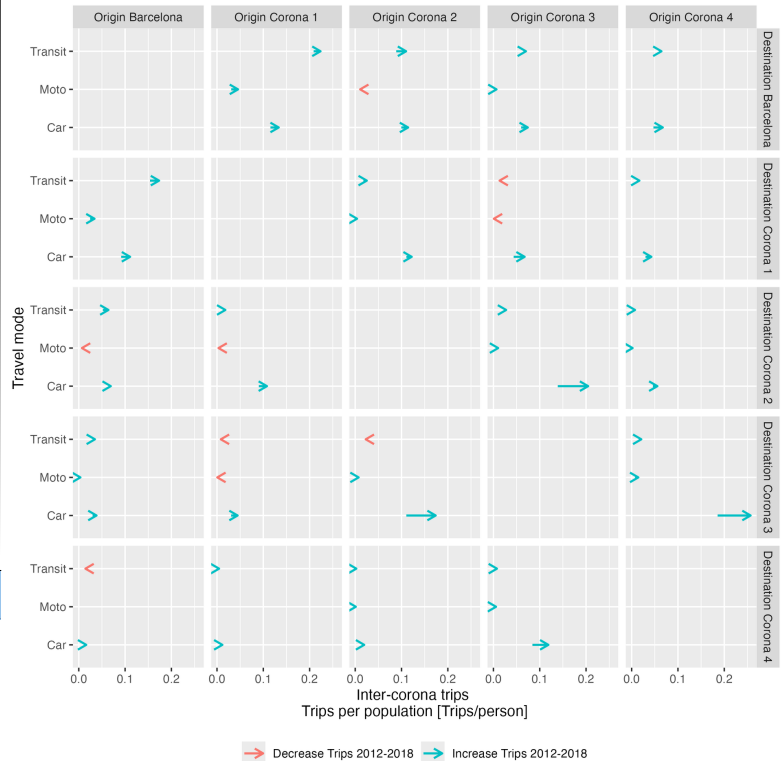
RESULTS/FINDINGS

- Decline in Intra-Corona Trips: Intra-corona trips decreased overall, especially in walking, even with a minor increase in car trips. This trend suggests that local accessibility alone may be insufficient to maintain high walking levels without enhanced regional transit support.
- Increase in Inter-Corona Trips: Car trips between coronas increased significantly, reflecting a regional trend toward motorized travel for longer distances. Limited regional transit options contribute to this shift.
- Transit supply has lagged behind demand, with minimal expansions and service reductions outside Barcelona's core, fueling increased car dependency

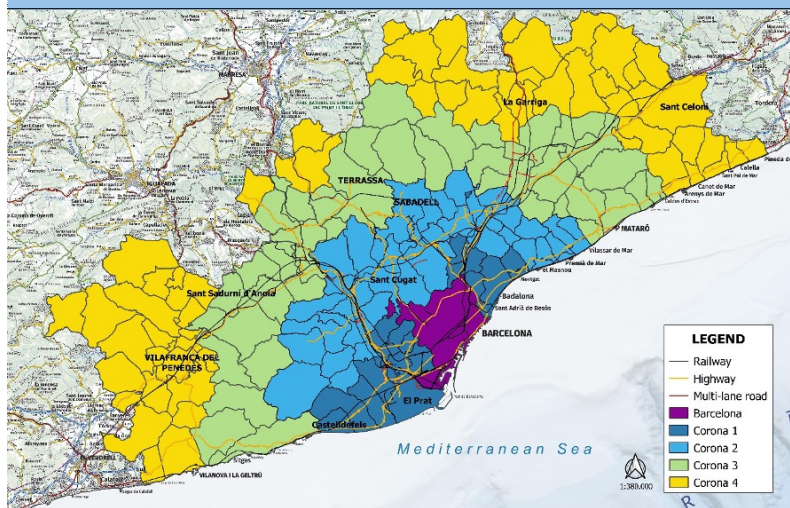
Intra-corona trips evolution by corona



Inter-corona trips evolution by corona



Transit fare coronas



CONCLUSION

The study reveals that in the peri-urban and suburban zones of Barcelona, shorter walking trips are increasingly replaced by car travel due to limited transit options and lower transit efficiency compared to car use. Inter-zonal car trips have grown substantially. To reduce car dependency, planners should prioritize enhancing both inter- and intra-corona transit connections across the metropolitan region